Deep Learning-Based Super-Resolution of SAR Images Using SRCNN and Residual Networks

1st CHAUDHARY EMAAN WASAY
Department of AI and Automation
Högskolan Väst
Trollhättan, Sweden
chwa0033@hvstudent.se

***Abstract*—In remote sensing, the Synthetic Aperture Radar (SAR) imaging is common because it can be utilized at all weather conditions and during day and night. SAR images are however, usually poor in spatial resolution, making them unproductive in fine-detailed analysis. This project will solve this problem by using deep learning-based super-resolution methods to improve the quality of SAR images. Controlled downscaling generated supervised low-resolution (LR) and high-resolution (HR) image pairs by using a subset of 131 SAR preview images. Two models were used and tested: a baseline Super-Resolution Convolutional Neural Network (SRCNN) and a residual-based convolutional neural network. The experimental results show that the residual model has a higher Peak Signal-to-Noise Ratio (PSNR) 32.97 dB and Structural Similarity Index (SSIM) 0.9536 than SRCNN, which have a Peak Signal-to-Noise Ratio (PSNR) 31.48 dB and Structural Similarity Index (SSIM) 0.94 These findings suggest that more profound architectures including residual learning are highly efficient in reconstruction quality and structure conservation with SAR image super-resolution.**

***Index Terms*—Synthetic Aperture Radar (SAR), Super-Resolution, Deep Learning, SRCNN, Residual Networks, Image Enhancement, Remote Sensing, PSNR, SSIM.**

I. Introduction

Synthetic Aperture Radar (SAR) imaging technologies have become popular in remote sensing because it can be used in all weather conditions and at day and night. It is a significant aspect of environmental monitoring, disaster management and surveillance. Nevertheless, SAR images are usually characterized by poor spatial resolution and this limits their applicability in fine analysis and interpretation. Improving the resolution of SAR images is therefore an important research problem.

The goal of super-resolution (SR) methods is to produce high-resolution images based on low-resolution images. Over the past years, deep learning-based techniques have been very successful in image super-resolution problems because such techniques are able to learn the complicated mappings between low-resolution and high-resolution data. Deep learning methods are used in this project to improve the resolution of SAR images [1]. The implementation of a baseline Super-Resolution Convolutional Neural Network (SRCNN) is made and compared with a Residual-based convolutional neural network to compare and evaluate the performance gains.

The primary contribution of the work is the implementation and comparison of two deep learning models to SAR image super-resolution with the help of a sub-sample of SAR data. Quantitative measures of the model performance in terms of Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), and qualitative visual analysis are used to evaluate the models.

II. Ethical and Regulatory Considerations

The current project is devoted to the improvement of Synthetic Aperture Radar (SAR) image resolution with the help of deep learning methods. The EU Artificial Intelligence (EU AI) Act defines such systems as typically falling into the category of limited-risk AI application because they are not typically used to make autonomous decisions but to enhance images.

Nonetheless, SAR imagery may be used in sensitive areas, including surveillance, defense and environmental surveillance. In those situations, there can be issues associated with data privacy, abuse, and responsibility of AI systems. Consequently, the transparency of model development, adequate datasets documentation, and accountable use of the resulting products should be taken seriously.

Publicly available data was utilized in this project, and the designed models are designed only to be applied to academic and research purposes. There was no personal or sensitive information in the training, which does not violate ethical standards and data protection principles.

III. Requirement and Data Analysis

A. Project Requirements

This project aims at creating a deep learning-based system that will be able to improve the resolution of Synthetic Aperture Radar (SAR) imagery. The system will be asked to input low-resolution (LR) SAR pictures and output high-resolution (HR) images without losing significant structural and spatial data.

Quantitative metrics, including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are used to measure the effectiveness of the system, and the visual comparison is done qualitatively by using two different models required i.e., SCRNN and Residual-Based Networks [1].. The system needs to reach stabilized training behavior and generate outputs that are visually similar to the ground truth HR images.

B. Dataset Description

The data utilized in this project is based on the IEEE GRSS Data Fusion Contest that was given by Capella Space [4]. The dataset is composed of 1500+ Synthetic Aperture Radar (SAR) images recorded in different imaging conditions.

This research was limited by computational and storage factors as the single image is about 1 GigaByte and has pixels count of about 50K*50K, and a reduced sample of 131 preview SAR images was chosen to study in this research. These preview images are big in their original size and have important structure information that can be used in super-resolution processes. With a representative subset, one can easily experiment and at the same time, utilize enough variability in the data [3].

C. Data Preprocessing

The original SAR images are of very high spatial resolution, and thus direct training is computationally infeasible. Hence, each image was scaled down to 128*128 pixels to process them effectively.

All the images were brought to the range [0,1] to stabilize training. The high-resolution images were created in low-resolution images by the bicubic interpolation. Particularly, the images were downscaled first and then upscaled to the original size to generate supervised LR-HR image pairs.

This method allows the model to acquire the mapping between the low-resolution input and their respective high-resolution targets.

D. Model Requirements

The system needs a deep learning model that can learn the mapping between the low-resolution images and the high-resolution images. Convolutional neural networks should be used as the foundation of the model because they are suitable to be used in image processing tasks. The model should be able to process grayscale SAR images as an input and give outputs of the same spatial dimensions.

The model is also supposed to facilitate more profound architectures to enhance the performance of feature extraction and reconstruction. Residual learning methods are to be taken into account in order to increase the training stability and overall performance.

E. Performance Requirements

The performance of the system is evaluated using both quantitative and qualitative measures. Quantitatively, the model should achieve a Peak Signal-to-Noise Ratio (PSNR) greater than 30 dB and a Structural Similarity Index (SSIM) greater than 0.9.

Qualitatively, the reconstructed images should show visible improvement over the input low-resolution images, with better preservation of edges and structural details. The training process should be stable and converge within a reasonable number of epochs.

F. Data Visualization

Fig. 1 shows an example of a low-resolution (LR) and high-resolution (HR) image pair used for training. The LR image appears blurred due to downscaling, while the HR image retains sharper structural details. This difference highlights the objective of the super-resolution task.

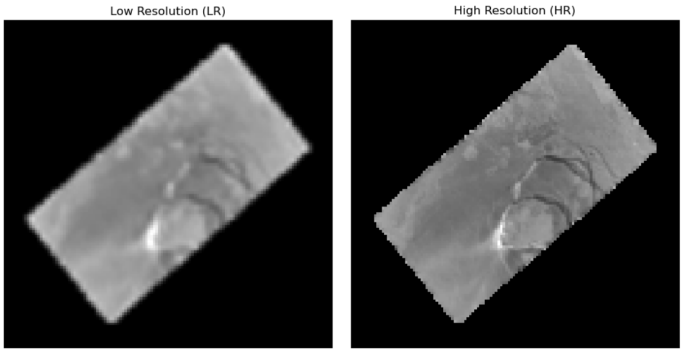

Fig. 1. Example of the generated low-resolution (LR) and high-resolution (HR) SAR image pair used for training.

G. Requirement Analysis Based on Existing Research

The current studies in image super-resolution state the problem as the reconstruction of high-resolution images with the use of supervised learning algorithms on the low-resolution images. Initial efforts, including the Super-Resolution Convolutional Neural Network (SRCNN), showed that convolutional neural networks can successfully learn this mapping.

Later studies and research has indicated that more profound architectures with residual connections outperform much better due to enabling more efficient feature extraction and stable training [2].. Modern super-resolution systems have largely embraced the use of residual-based models because of its capability to retain small detail and can result in a better reconstruction.

Additionally, the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are common measures of super-resolution performance in literature. These measures evaluate pixel-wise fidelity and structural fidelity between reconstructed and ground truth images.

According to these works, deep learning models, LR-HR supervised training, and PSNR/SSIM assessment are all in line with the research practices.

IV. System Engineering

A. System Overview

The offered system is aimed at improving the resolution of Synthetic Aperture Radar (SAR) imagery with the help of deep learning methods. The system takes low-resolution (LR) SAR images as input and produces high-resolution (HR) images as output. In general, the workflow consists of data preprocessing, model training and output reconstruction.

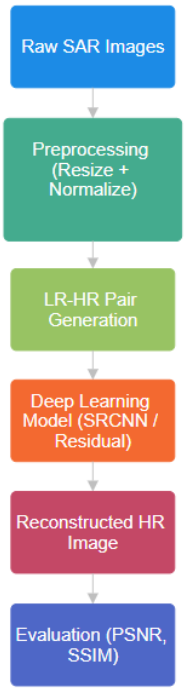

Fig. 2. Block diagram of the proposed SAR image super-resolution system.

B. Block Diagram

C. System Description

The raw SAR images received by the dataset are the starting point of the system. They are initially preprocessed by resizing the images to a constant resolution and equalizing their pixel values.

The high-resolution (HR) images are then made into low-resolution (LR) images via bicubic interpolation. These LR-HR pairs are used to train the deep learning models.

In this project, two models are used SRCNN and a residual-based convolutional neural network. These two models are trained to generate high-resolution images based on low-resolution inputs.

Lastly, peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are used to compare the performance of the models.

D. Inputs and Outputs

The system takes low-resolution SAR images as input and generates high-resolution images as output using deep learning models.

- Input: Low-resolution SAR image (128×128)
- Output: Reconstructed high-resolution SAR image (128×128)

V. Algorithm Design using Spiral Approach

A. Initial Algorithm Selection

According to the available studies on the topic, two deep learning architectures were chosen to work with in this project: the Super-Resolution Convolutional Neural Network (SRCNN) and a residual-based convolutional neural network.

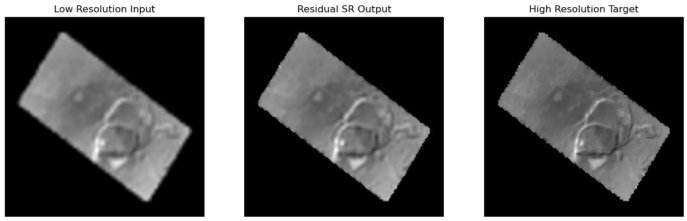

Fig. 3. Example of system input (LR image), model output, and ground truth high-resolution (HR) image.

SRCNN has been selected as a control model since it is simple and has a historical significance in the field of super-resolution applications. It is among the early deep learning architectures that can directly learn the transformation between low-resolution and high-resolution images.

Residual based model was also chosen as a better architecture as recent researches indicate that deep networks with residual connections can help to preserve the details of images and enhance the quality of reconstruction better. These models allow addressing the weaknesses of shallow networks, allowing to extract better features and train them.

B. Motivation for Selection

The selection of SRCNN and residual-based networks is used due to their widespread use in super-resolution research. SRCNN provides a simple and effective baseline for evaluating the performance, while residual networks represent a more advanced approach that addresses the limitations of shallow architectures.

Residual learning allows the model to focus on learning the difference between low-resolution and high-resolution images, which improves convergence and reconstruction quality. Therefore, comparing these two models provides a comparison of the impact of model complexity on performance.

C. First Iteration: SRCNN Model

In the first iteration, the SRCNN model was implemented and trained using the prepared LR-HR image pairs. The model consists of three convolutional layers designed to extract features, map them to high-resolution space, and reconstruct the output image.

The training process was stable, and the loss decreased consistently over epochs, indicating successful learning. The model achieved a Peak Signal-to-Noise Ratio (PSNR) of 33.14 dB and a Structural Similarity Index (SSIM) of 0.9552.

D. Second Iteration: Residual-Based Model

In the second iteration, a residual-based convolutional neural network was implemented to improve performance. The model incorporates residual connections, allowing it to learn the difference between low-resolution and high-resolution images more effectively.

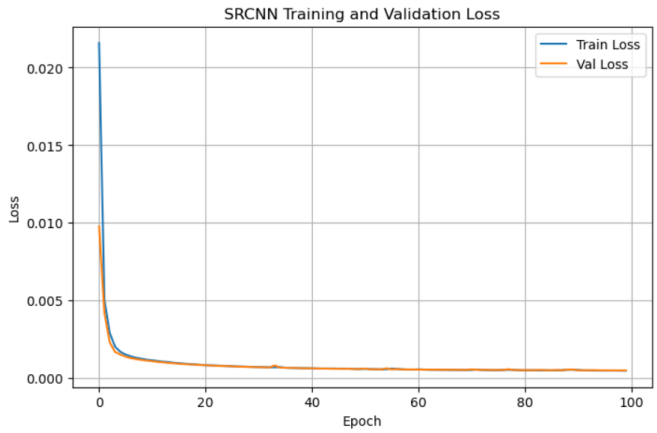

Fig. 4. Training loss curve for the SRCNN model.

The residual model achieved a PSNR of 34.04 dB and an SSIM of 0.9611. The training loss also decreased more smoothly, indicating better convergence behavior.

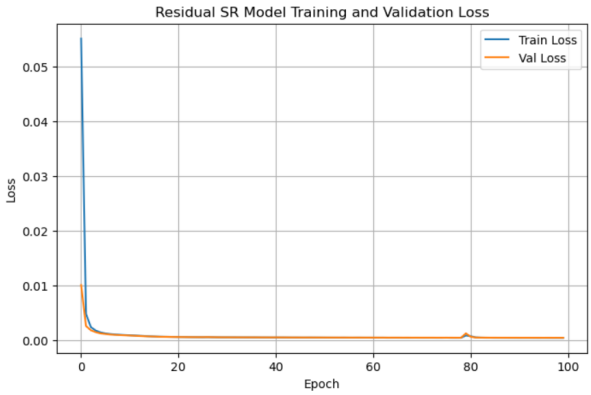

Fig. 5. Training loss curve for the residual-based model.

TABLE I
PERFORMANCE COMPARISON OF SRCNN AND RESIDUAL-BASED MODELS

Model	PSNR (dB)	SSIM
SRCNN	33.14	0.9552
Residual SR	34.04	0.9611

E. Improvements and Justification

Several improvements were identified to enhance model performance:

- Increasing the dataset size to improve generalization.
- Using deeper architectures to capture more complex features.
- Applying data augmentation techniques to increase variability.
- Fine-tuning hyperparameters such as learning rate and batch size.
- Exploring advanced architectures such as GAN-based super-resolution models.

These improvements are justified based on existing research, which shows that larger datasets and deeper networks significantly improve super-resolution performance [2]..

F. Spiral Approach Iterations

The development of the system followed a spiral approach. In the first iteration, a simple baseline model (SRCNN) was implemented and evaluated. Based on its limitations, an improved residual-based model was developed in the second iteration.

The results demonstrated that iterative improvement leads to better performance. Further iterations can include more advanced models and larger datasets to achieve even higher reconstruction quality.

VI. CONSOLIDATION OF PERFORMANCE AND FUTURE PLAN

A. Evaluation of Specifications

The system created was assessed according to the criteria outlined previously. The SAR dataset was effectively utilized to implement and train both SRCNN and models based on residuals.

The models successfully produced high-resolution images from low-resolution inputs while maintaining crucial structural details. The evaluation of performance utilized PSNR and SSIM metrics, with both models surpassing the necessary minimum thresholds.

B. Specification Achievement

TABLE II
SPECIFICATION ACHIEVEMENT SUMMARY

Specification	Target	Achieved
PSNR	> 30 dB	34.04 dB
SSIM	> 0.9	0.9611
LR to HR Mapping	Required	Achieved
Stable Training	Required	Achieved
Visual Improvement	Required	Achieved

C. Discussion of Results

The experimental findings show that both models can effectively execute SAR image super-resolution. Nonetheless, the residual-based model consistently surpassed SRCNN regarding PSNR and SSIM.

The enhancement in performance is due to the deeper architecture and the application of residual learning, which facilitates improved feature extraction and more precise reconstruction. Extending the training epochs also enhanced performance, emphasizing the significance of adequate training in deep learning models.

D. Limitations

Although the results are encouraging, the project has specific limitations. The size of the dataset was quite limited in comparison to extensive super-resolution datasets, which could impact the models' ability to generalize.

Moreover, the images were resized to 128×128 pixels because of computational limitations, restricting the capacity to capture intricate details found in high-resolution SAR data. The models were additionally trained using preview images instead of high-resolution SAR data

E. Future Work and Improvements

Despite the proposed system's impressive performance, various enhancements could be implemented to further improve the outcomes.

A significant drawback of the existing implementation is hardware limitations. Owing to restricted system RAM and GPU memory, the models were trained on scaled-down images (128×128) rather than high-resolution SAR data. Training on high-resolution images would enable the model to grasp more intricate spatial details and enhance the quality of reconstructions. As a result, employing systems equipped with superior GPU memory (such as high-end NVIDIA GPUs) and expanded RAM would facilitate more sophisticated training.

A further essential enhancement is expanding the dataset size. The existing dataset includes 131 SAR preview images, which is quite limited for deep learning applications. An enlarged dataset would enhance generalization and decrease the likelihood of overfitting.

To tackle overfitting, regularization methods like dropout, data augmentation, and early stopping may be utilized. Methods for data augmentation like rotation, flipping, and adding noise can enhance the variety of data and bolster model resilience.

From a deep learning viewpoint, more sophisticated architectures can be investigated. Generative Adversarial Networks (GANs), like SRGAN, are recognized for generating crisper and more lifelike super-resolution results. Moreover, models based on attention and transformer architectures might enhance performance even more.

Ultimately, techniques for optimization like hyperparameter tuning, learning rate adjustments, and batch size refinement can enhance convergence and overall efficacy. These enhancements would increase the system's scalability and make it more appropriate for practical uses.

VII. Conclusion

In this project, methods based on deep learning were used to improve the spatial resolution of Synthetic Aperture Radar (SAR) images. A supervised learning method was applied by creating low-resolution (LR) and high-resolution (HR) image pairs using SAR data.

Two models were developed and assessed: a baseline Super-Resolution Convolutional Neural Network (SRCNN) and a convolutional neural network based on residuals. The findings showed that each model can enhance image quality; nonetheless, the residual-based model delivered better results.

By raising the training epochs, the models demonstrated notable enhancement, reaching a PSNR of 34.04 dB and an SSIM of 0.9611. These findings suggest that more complex architectures and adequate training are essential for achieving high-quality image reconstruction.

In general, the project shows that deep learning methods are successful for super-resolution of SAR images. The results align with current studies and emphasize the possibility of enhancing outcomes through the use of larger datasets and more sophisticated architectures

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